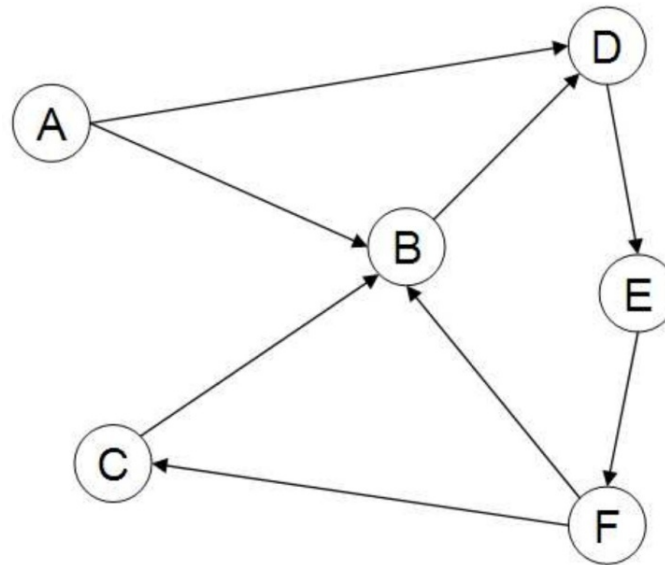

Information Retrieval

Ranking: PageRank

- Link structure provides information to measure 'importance' of a page
- Web = directed graph



- Inlinks: Pages that contain links to a page (in-degree)
 - ✓ On average between 8 to 15 (for entire web)
- Outlinks: Other pages that a page contains links to (out-degree)

Ranking: PageRank

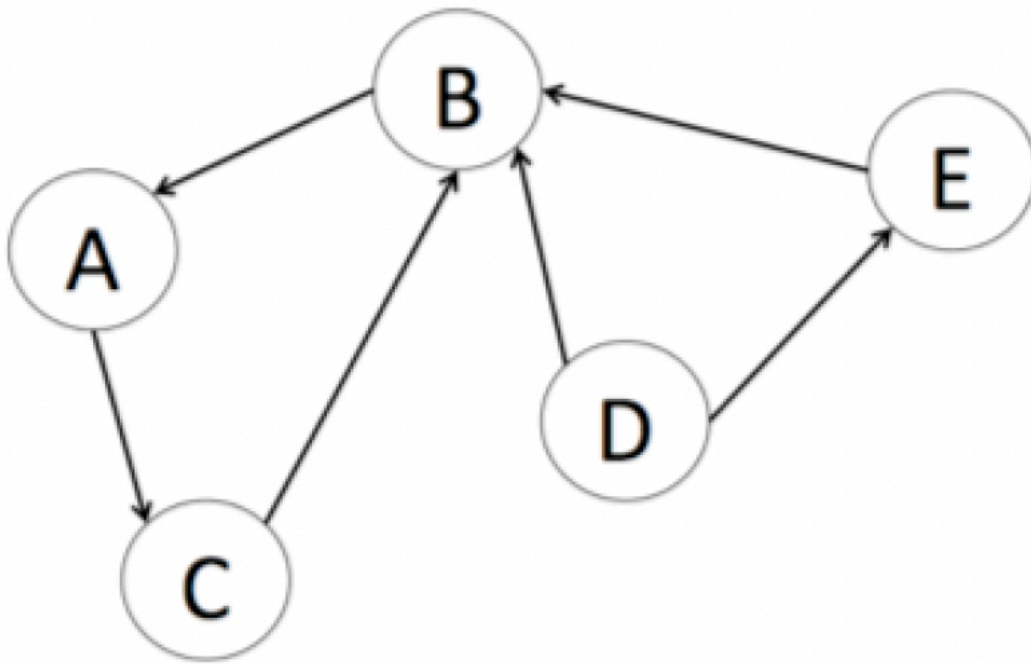
- Basis for [Google's PageRank algorithm](#) (Brin & Page 1998):
- Page rank PR_i of a page i increases with the number and page rank of the pages linking to it.

$$PR_i = \frac{1-d}{N} + d \sum_{\forall j \in \{(j,i)\}} \frac{PR_j}{C_j}$$

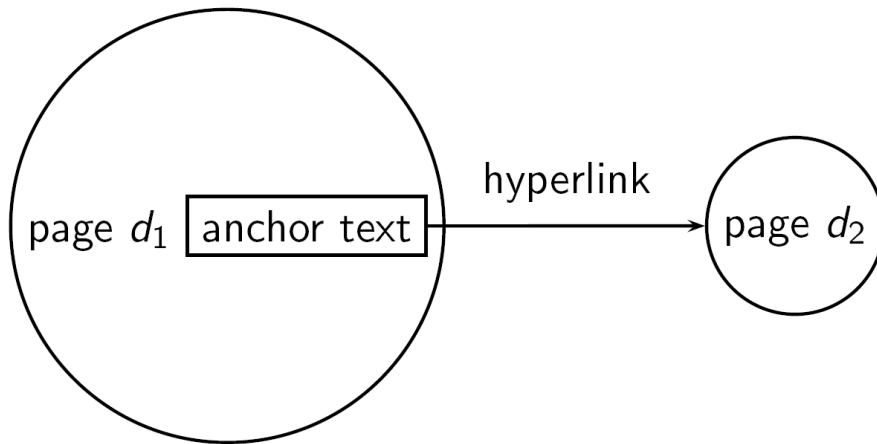
- d = dampening factor, N = total number of pages, C_j = number of outlinks of page j
- Recursive definition, Brin & Page (1998) describe iterative algorithm
 - About 100 iterations required for entire web

Exercise

Compute the page ranks (damping factor = 0.8) for nodes in the graph in Figure. Initialize with a PageRank value of 0.2.



Ranking: Anchor texts



Example:

“You can find cheap cars [here](#).”

Anchor text (loose definition):

“You can find cheap cars here”

Anchor text (strict definition):

“here”

Assumption 1: A hyperlink is a quality signal.

The hyperlink $d_1 \rightarrow d_2$ indicates that d_1 's author deems d_2 relevant.

Assumption 2: The anchor text describes the content of d_2 .

We use anchor text somewhat loosely here for: the text surrounding the hyperlink.

Ranking: Anchor texts

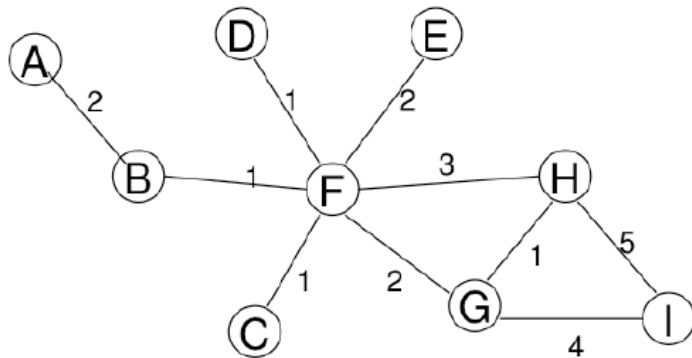
- Searching on [text of document + anchor text → document] is often more effective than searching on [text of document only].
- Example:
 - For *ibm* how to distinguish between:
 - IBM's home page (mostly graphical)
 - *May not even match the query!*
 - IBM's copyright page (high term freq. for 'ibm')
 - Rival's spam page (arbitrarily high term freq.)
- Thus: Anchor text is often a better description of a page's content than the page itself.
 - Anchor text can be weighted more highly than document text. (based on Assumptions 1 & 2)

Graphs

Why Graph?

- ✓ Graphs are an intuitive and natural way to encode language units as nodes and their similarities as edges
 - feature-based representation can be transformed into a graph via a similarity measure
- ✓ There exist methods that directly operate on graphs
 - graph clustering, e.g. kNN
 - graph-based ranking, e.g. PageRank
 - path-based algorithms, e.g. Dijkstra

Graph Representations



Geometrical representation

Set representation

$G(V, E)$

$V = \{A, B, C, D, E, F, G, H, I\}$

$E = \{AB, BF, CF, DF, EF, FG, FH, GH, GI, HI\}$

$ew(BF) = ew(CF) = ew(DF) = ew(GH) = 1;$

$ew(AB) = ew(EF) = ew(FG) = 2; ew(FH) = 3;$

$ew(GI) = 4; ew(HI) = 5;$

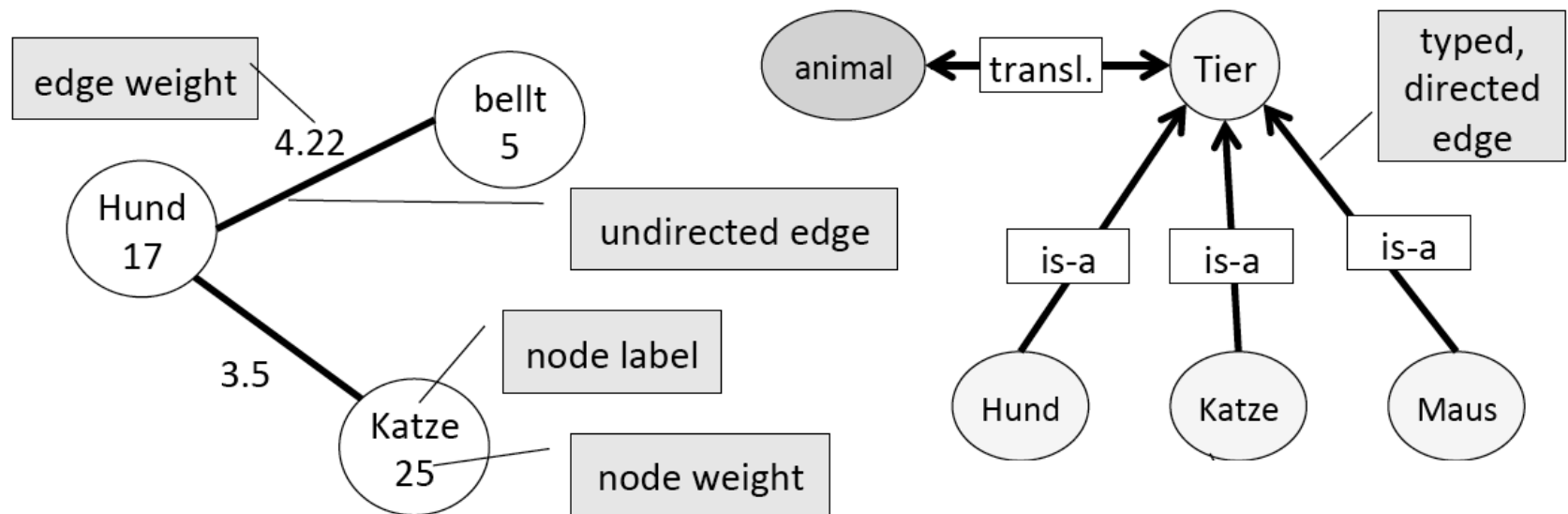
a_{ij}	A	B	C	D	E	F	G	H	I
A	0	2	0	0	0	0	0	0	0
B	2	0	0	0	0	1	0	0	0
C	0	0	0	0	0	1	0	0	0
D	0	0	0	0	0	1	0	0	0
E	0	0	0	0	0	2	0	0	0
F	0	1	1	1	2	0	2	3	0
G	0	0	0	0	0	2	0	1	4
H	0	0	0	0	0	3	1	0	5
I	0	0	0	0	0	0	4	5	0

Adjacency matrix

Graphs in NLP

- ✓ Graph $G(V,E)$ with V = set of nodes, E = set of edges
 - ✓ nodes can be (multi-) labelled, weighted
 - ✓ edges can be weighted, typed, directed or undirected
- ✓ Graphs encode relationships between entities, which are represented as nodes

Graphs in NLP



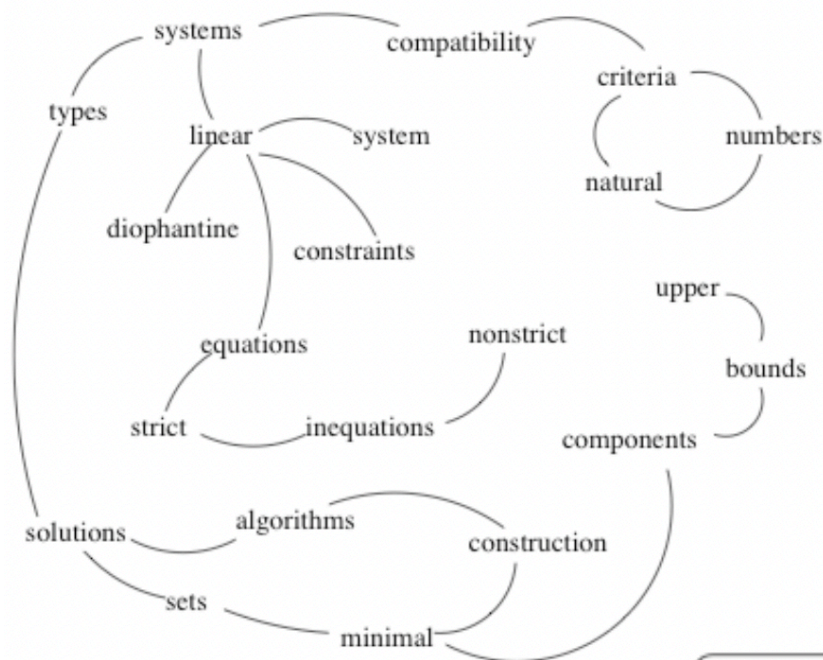
Co-occurrence Graphs

- *If two words X and Y occur together in some contextual unit of information (as neighbours, in a word window of 5, in a clause, in a sentence, in a paragraph), they are said to co-occur.*
- *Words as vertices and edge weights as the number of times two words co-occur.*

TextRank for Keyword Extraction

- *Keyword extraction: find the most salient keywords for a document*
- *Keyword extraction with PageRank:*
 - *preprocess document: identify adjectives and nouns as targets*
 - *target co-occurrence graph: targets co-occurring within a window of 2-10 words*
 - *apply PageRank to get ranking scores on nodes*
 - *select highest scoring keywords*

TextRank for Keyword Extraction



Keywords assigned by TextRank:

linear constraints; linear diophantine equations; natural numbers; nonstrict inequations; strict inequations; upper bounds

Keywords assigned by human annotators:

linear constraints; linear diophantine equations; minimal generating sets; non-strict inequations; set of natural numbers; strict inequations; upper bounds

Lex Rank for Summarization

	TF: count $w_1..w_n$					TF*IDF				
Sentence	w1	w2	w3	...	wn	w1	w2	w3	...	wn
This is a sentence that talks about some topic.	3	0	2		0	0	0	.27		0
And here is another sentence that talks abot something slightly different.	5	0	3		1	0	0	.24		.21
...				...					...	
And here is yet another one of these notorious sentences	7	4	0		0	0	.62	0		0
DF	3	1	2	...	1	$IDF(w) = \log\left(\frac{\text{total number of sentences}}{DF(w)}\right)$				

feature vector of the second sentence

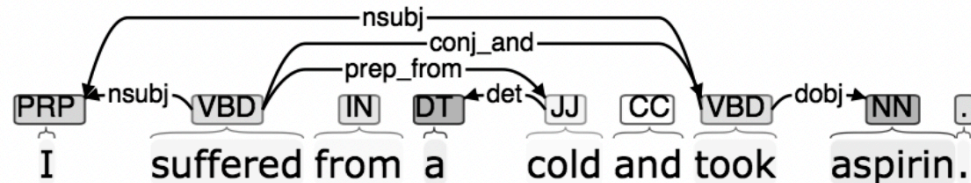
This is the same as the vector space model for Information Retrieval

Lex Rank for Summarization

- *Sentence similarity graph:*
 - *nodes: sentences*
 - *edges: cosine similarity between sentence feature vectors*
- *Can apply a threshold on similarity*

Distributional Thesaurus Network

SENTENCE:



STANFORD COLLAPSED DEPENDENCIES:

<http://nlp.stanford.edu:8080/parser/>

nsubj(suffered, I); nsubj(took, I); root(ROOT, suffered); det(cold, a);
prep_from(suffered, cold); conj_and(suffered, took); dobj(took, aspirin)

TERM-CONTEXT PAIRS:

suffered	nsubj(@@, I)	1
took	nsubj(@@, I)	1
cold	det(@@, a)	1
suffered	prep_from(@@, cold)	1
suffered	conj_and(@@, took)	1
took	dobj(@@, aspirin)	1

I	nsubj(suffered, @@)	1
I	nsubj(took, @@)	1
a	det(cold, @@)	1
cold	prep_from(suffered, @@)	1
took	conj_and(suffered, @@)	1
aspirin	dobj(took, @@)	1

Distributional Thesaurus Network

- *Computed from distributional similarity statistics*
- *Entry for a target word consists of a ranked list of neighbours*

meeting	288	articulate	89
meetings	102	explain	19
hearing	89	understand	17
session	68	communicate	17
conference	62	defend	16
summit	51	establish	15
forum	46	deliver	14
workshop	46	evaluate	14
hearings	46	adjust	14
ceremony	45	manage	13
sessions	41	speak	13
briefing	40	change	13
event	40	answer	13
convention	38	maintain	13
gathering	36	...	
...		...	

